Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics Practical

Practical – VII

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Practical 7: Sampling and Sampling Distributions

Part A: Survey and Sampling

1. Select a real-life business survey topic.
2. Prepare at least 5 survey questions.
3. Collect responses from 50+ people.
4. Store the data in CSV/Pandas DataFrame.
5. Display the first 5 records.
6. Identify the population and sample.
7. Identify the sampling technique.
8. Identify numerical and categorical variables.
9. Select one numerical variable.
10. Calculate sample size, mean, median and standard deviation.
11. Create a suitable chart.
12. Write two observations.

Code :

```
# PART A - Survey and Sample Data Analysis
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load the Anti-Ragging Awareness CSV dataset
df = pd.read_csv("Anti_Ragging_Awareness_CEP_Dataset.csv")
print("Dataset Loaded Successfully!")
print("Total Records:", len(df))
print("Total Columns:", len(df.columns))
# Display first five records
display(df.head())
# Identify population
print("\nPopulation:")
print("All people in the target community who could participate in the Anti-Ragging
Awareness survey.")
# Identify sample
print("\nSample:")
print("75 respondents included in the Anti-Ragging Awareness survey dataset.")
# Identify sampling technique
print("\nSampling Technique:")
print("Convenience Sampling")
# Identify numerical variables
numerical_variables = [
    "Seriousness Rating",
    "Information Useful",
    "Awareness Increased"
]
print("\nNumerical Variables:")
for variable in numerical_variables:
    print(variable)
# Identify categorical variables
categorical_variables = [
    "Name",
    "Age Group",
    "Gender",
    "Occupation",
    "Education Level",
```

```
        "Location",
        "Heard About Ragging",
        "Knows What Is Ragging",
        "Knows It Is Prohibited",
        "Knows Reporting Method",
        "Knows Helpline",
        "Witnessed/Experienced",
        "Awareness Programs Helpful",
        "Will Share Information",
        "Feedback"
    ]
    print("\nCategorical Variables:")
    for variable in categorical_variables:
        print(variable)
    # Select numerical variable for analysis
    selected_variable = "Awareness Increased"
    # Remove missing values
    data = df[selected_variable].dropna()
    # Calculate sample size
    sample_size = data.count()
    # Calculate mean
    sample_mean = data.mean()
    # Calculate median
    sample_median = data.median()
    # Calculate standard deviation
    sample_std = data.std()
    print("\nSelected Variable:", selected_variable)
    print("Sample Size:", sample_size)
    print("Mean:", round(sample_mean, 2))
    print("Median:", sample_median)
    print("Standard Deviation:", round(sample_std, 2))
    # Create histogram
    plt.figure(figsize=(8, 5))
    plt.hist(
        data,
        bins=np.arange(0.5, 6.5, 1),
        edgecolor="black"
    )
    plt.title("Awareness Increased Among Respondents")
    plt.xlabel("Awareness Increased Rating")
    plt.ylabel("Number of Respondents")
    plt.xticks([1, 2, 3, 4, 5])
    plt.grid(axis="y", alpha=0.3)
    plt.show()
    # Write two observations
    print("\nObservations:")
    print("1. Most respondents reported a moderate to high increase in awareness.")
    print("2. The survey shows that the awareness activity helped improve understanding
    about anti-ragging.")
```

Output :

```
Dataset Loaded Successfully!
Total Records: 75
Total Columns: 19
```

	Timestamp	Name	Age Group	Gender	Occupation	Education Level	Location	Heard About Ragging	Knows What Is Ragging	Knows It Is Prohibited	Knows Reporting Method	Knows Helpline	Witnessed/Experienced	Seriousness Rating	Awareness Programs Helpful	Information Useful	Awareness Increased	Will Share Information	Feedback
0	20-09-2026 09:00:00	Aditya Patil	19-21	Male	Student	Secondary School	Andheri, Mumbai	Yes	Yes	Yes	No	No	Yes	3	Yes	5	5	Yes	Very informative session.
1	20-09-2026 09:37:00	Aarav Shinde	19-21	Male	Student	Secondary School	Andheri, Mumbai	Yes	Yes	Yes	Yes	Yes	Yes	5	Yes	5	5	Yes	Helpful and easy to understand.
2	20-09-2026 10:14:00	Omkar Jadhav	13-15	Male	Student	Secondary School	Andheri, Mumbai	Yes	Yes	Yes	Yes	Yes	No	3	Maybe	5	5	Yes	Helpful and easy to understand.
3	20-09-2026 10:51:00	Siddhant Pawar	19-21	Male	Student	Secondary School	Andheri, Mumbai	Yes	Yes	Yes	Yes	Yes	No	3	Yes	4	5	Yes	Good initiative for students.
4	20-09-2026 11:28:00	Rohan More	19-21	Male	Student	Higher Secondary	Andheri, Mumbai	Yes	Yes	Yes	No	No	No	4	Yes	3	4	Yes	Helpful and easy to understand.

```
Population:
All people in the target community who could participate in the Anti-Ragging Awareness survey.

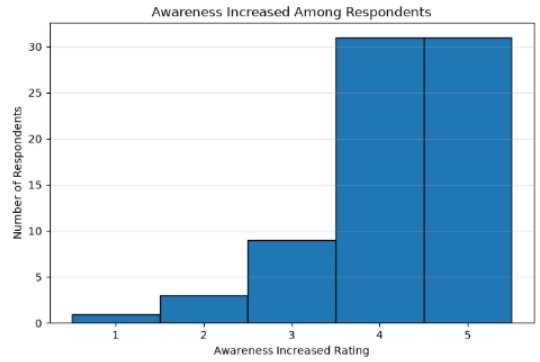
Sample:
75 respondents included in the Anti-Ragging Awareness survey dataset.

Sampling Technique:
Convenience Sampling

Numerical Variables:
Seriousness Rating
Information Useful
Awareness Increased

Categorical Variables:
Name
Age Group
Gender
Occupation
Education Level
Location
Heard About Ragging
Knows What Is Ragging
Knows It Is Prohibited
...
Sample Size: 75
Mean: 4.17
Median: 4.0
Standard Deviation: 0.89
```

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```
Observations:
1. Most respondents reported a moderate to high increase in awareness.
2. The survey shows that the awareness activity helped improve understanding about anti-ragging.
```

Part B: Sampling Distribution and CLT

13. Select a numerical variable.
14. Randomly select 30 observations.
15. Calculate the sample mean.
16. Repeat sampling 100 times.
17. Calculate and store each sample mean.
18. Display the first 10 sample means.
19. Calculate the mean of sample means.
20. Calculate the standard deviation of sample means.
21. Create a histogram of sample means.
22. Observe the distribution shape.
23. Compare the original mean with the sampling mean.
24. Compare distributions for different sample sizes.
25. Calculate the standard error.
26. Explain the Central Limit Theorem.
27. Estimate the population mean using the sample mean.
28. Write the conclusion.

Code :

```
# PART B - Sampling Distribution
# Select numerical variable
selected_variable = "Awareness Increased"
# Remove missing values
data = df[selected_variable].dropna()
# Select a random sample of 30 observations
random_sample_30 = data.sample(
    n=30,
    random_state=42
)
# Calculate sample mean
sample_30_mean = random_sample_30.mean()
print("\nRandom Sample of 30 Observations:")
print(random_sample_30.values)
print("\nMean of Selected Sample of 30:")
print(round(sample_30_mean, 2))
# Repeat random sampling 100 times
sample_means_30 = []
for i in range(100):
    sample = data.sample(
        n=30,
        replace=True
    )
    sample_mean_value = sample.mean()
    sample_means_30.append(sample_mean_value)

print("\n100 Random Samples Generated Successfully.")

# Store sample means in a DataFrame
```

```
sampling_distribution = pd.DataFrame({
    "Sample Number": range(1, 101),
    "Sample Mean": sample_means_30
})
# Display first 10 sample means
print("\nFirst 10 Sample Means:")
display(sampling_distribution.head(10))
# Calculate mean of all sample means
mean_of_sample_means = pd.Series(sample_means_30).mean()
print("\nMean of All Sample Means:")
print(round(mean_of_sample_means, 2))
# Calculate standard deviation of sample means
std_of_sample_means = pd.Series(sample_means_30).std()
print("\nStandard Deviation of Sample Means:")
print(round(std_of_sample_means, 2))
# Create histogram of sampling distribution
plt.figure(figsize=(8, 5))
plt.hist(
    sample_means_30,
    bins=10,
    edgecolor="black"
)
plt.axvline(
    mean_of_sample_means,
    linestyle="--",
    linewidth=2,
    label="Mean of Sample Means"
)
plt.title("Sampling Distribution of Sample Means (n = 30)")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.grid(axis="y", alpha=0.3)
plt.show()
# Observe the shape
print("\nShape of Sampling Distribution:")
print("The sampling distribution is approximately bell-shaped and concentrated around
its mean.")
# Compare original mean and sampling distribution mean
original_mean = data.mean()
print("\nComparison of Means:")
print("Original Dataset Mean:", round(original_mean, 2))
print("Mean of Sampling Distribution:", round(mean_of_sample_means, 2))
difference = abs(
    original_mean - mean_of_sample_means
)
print("Difference:", round(difference, 2))
# Compare sampling distributions for different sample sizes
sample_sizes = [10, 30, 50]
sampling_results = {}
for size in sample_sizes:
    means = []
    # Generate 100 samples for each sample size
    for i in range(100):

        sample = data.sample(
            n=size,
            replace=True
        )
        means.append(sample.mean())
    sampling_results[size] = means
    means_series = pd.Series(means)
    print(
        "Sample Size:", size,
        "| Mean:", round(means_series.mean(), 2),
        "| Standard Deviation:", round(means_series.std(), 2)
    )
# Plot sampling distributions for different sample sizes
plt.figure(figsize=(9, 6))
for size in sample_sizes:
    plt.hist(
        sampling_results[size],
        bins=10,
        alpha=0.5,
```

```
        label="Sample Size " + str(size)
    )
plt.title("Sampling Distributions for Different Sample Sizes")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.grid(axis="y", alpha=0.3)
plt.show()
# Calculate standard error
sample_std = data.std()
standard_error = sample_std / np.sqrt(30)
print("\nStandard Error:")
print(round(standard_error, 4))
# Explain Central Limit Theorem
print("\nCentral Limit Theorem:")
print(
    "The Central Limit Theorem states that when random samples "
    "are repeatedly taken, the distribution of their sample means "
    "becomes approximately normal as the sample size increases."
)
# Estimate population mean using sample mean
estimated_population_mean = sample_30_mean
print("\nEstimated Population Mean:")
print(round(estimated_population_mean, 2))
# Final conclusion
print("\nConclusion:")
print(
    "The Anti-Ragging Awareness survey used 75 respondents. "
    "Random samples of 30 observations were taken 100 times "
    "and their means were calculated. The sampling distribution "
    "was concentrated around the original mean. Larger sample "
    "sizes produced more stable sample means. This demonstrates "
    "the Central Limit Theorem and shows that a sample mean can "
    "be used to estimate the population mean."
)
```

Output :

```
Random Sample of 30 Observations:
[4 1 3 5 5 5 5 5 4 4 4 4 5 4 3 5 3 2 4 5 4 4 5 3 4 4 4 4 4 4]

Mean of Selected Sample of 30:
4.0

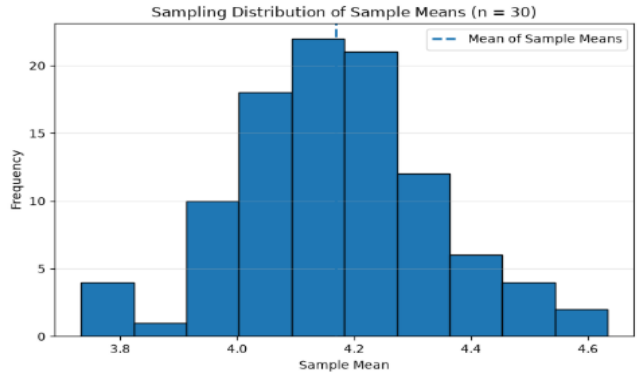
100 Random Samples Generated Successfully.

First 10 Sample Means:
```

	Sample Number	Sample Mean
0	1	4.233333
1	2	4.100000
2	3	4.033333
3	4	4.200000
4	5	4.333333
5	6	3.733333
6	7	4.033333
7	8	4.266667
8	9	4.466667
9	10	4.066667

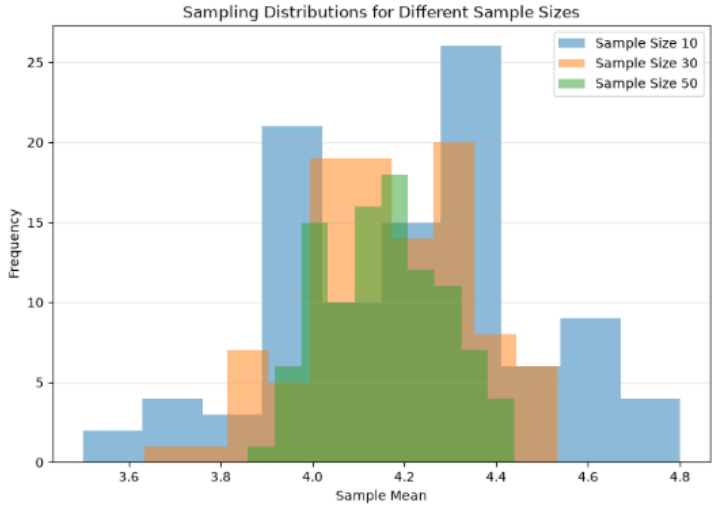
```
Mean of All Sample Means:
4.17

Standard Deviation of Sample Means:
0.17
```



```
Shape of Sampling Distribution:
The sampling distribution is approximately bell-shaped and concentrated around its mean.

Comparison of Means:
Original Dataset Mean: 4.17
Mean of Sampling Distribution: 4.17
Difference: 0.0
Sample Size: 10 | Mean: 4.2 | Standard Deviation: 0.27
Sample Size: 30 | Mean: 4.16 | Standard Deviation: 0.18
Sample Size: 50 | Mean: 4.16 | Standard Deviation: 0.13
```




```
Standard Error:
0.1627

Central Limit Theorem:
The Central Limit Theorem states that when random samples are repeatedly taken, the distribution of their sample means becomes approximately normal as the sample

Estimated Population Mean:
4.0

Conclusion:
The Anti-Ragging Awareness survey used 75 respondents. Random samples of 30 observations were taken 100 times and their means were calculated. The sampling distr
```